

SIH 2026 · ISRO · PS26171 · REPLAN

# 5-Day Sprint to 80% Functional

Real deadline: **Tuesday, 1 September 2026** — 5 working days from today (Thu 27 Aug). This replaces the earlier Sept-20 / December-paced plan; that timeline no longer applies here.

5 working days left

Target: 80% functional

Deadline: Tue 1 Sep 2026

6 members, 2 powerful laptops

## 01 What “80% functional” means — say the bar out loud

Building the full architecture as originally spec'd — in-browser vision model, full 3-tier privacy guard, PVM memory, cross-browser support, full benchmark suite — in 5 days with 6 people is not realistic, and pretending otherwise is how a team ends up with six unfinished pieces on Sept 1 instead of one working one. The honest move is to cut scope on purpose, not by accident.

**In scope — this is the 80%, must work live on the mock site:**

- Extension loads and captures DOM state of the mock test site.
- Tier-1 DOM-rule PII detection + deterministic redaction (blackout/mask) runs before any network call.
- Sanitized payload goes to a **cloud** LLM/VLM API; server returns one structured action (click/type/scroll) via a frozen JSON schema.
- Extension validates and executes that action locally.
- Level-1 deterministic verification (URL/DOM check) confirms the action actually worked.
- One canary-value test proves zero raw PII reached the server — the privacy proof.
- A basic privacy inspector panel shows what was redacted and what was sent.
- One complete task runs start-to-finish, repeatably, on the mock site.

**Deferred past Sept 1 — the hardest 20%, cut on purpose:**

- The in-browser local vision model (ONNX Runtime Web + WebGPU) — normally the multi-week piece on its own. Perception is **DOM/A11y-only** for this sprint.
- PVM verified-memory caching.
- Multi-site testing and Firefox compatibility.
- The full 5-metric benchmark suite.
- Visual/face redaction — text and DOM-field PII only for now.

## 02 Hardware — lighter load this week

Deferring the local vision model means Perception doesn't need standing GPU priority this sprint. Reallocate: **Server AI** gets first claim (fastest iteration on the cloud-API integration), **Integration** gets the second machine for wiring and the golden-path stress test. Keep both laptops free of anything that can run on a normal machine — every role below is scoped to not need one.

## 03 Day-by-day, all six roles

| Role       | Fri 28 Aug                                   | Sat 29 Aug                                      | Sun 30 Aug                                    | Mon 31 Aug                                | Tue 1 Sep                         |
|------------|----------------------------------------------|-------------------------------------------------|-----------------------------------------------|-------------------------------------------|-----------------------------------|
| Extension  | MV3 skeleton + content script into mock site | Action executor (click/type/scroll) + validator | type_secret pattern + error handling          | Full-team integration: fix breakages live | Bug triage only — no new features |
| Perception | DOM/A11y capture, normalize page state       | Hand off state to Privacy Guard + Server AI     | Polish capture on mock site's specific fields | Support integration + stress-test capture | Bug triage only                   |

| Role                 | Fri 28 Aug                                              | Sat 29 Aug                                                   | Sun 30 Aug                                           | Mon 31 Aug                                          | Tue 1 Sep                          |
|----------------------|---------------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------|-----------------------------------------------------|------------------------------------|
| <b>Privacy Guard</b> | Tier-1 DOM-rule detection + deterministic redaction     | Redaction validator + sanitized payload builder              | Canary-test harness + Privacy Firewall block-on-fail | Run canary test, capture SHA-256 proof              | Bug triage only                    |
| <b>Server AI</b>     | FastAPI skeleton; freeze action schema w/ Extension     | Sanitized-context endpoint live; hardcoded action to unblock | Real LLM reasoning returns valid structured actions  | Support integration + canary proof from server side | Bug triage only                    |
| <b>PVM/Verify</b>    | Stub only — help Integration build the mock site        | Level-1 deterministic verification (URL/DOM check)           | Basic recovery loop for failed actions               | Stress-test the one golden task for reliability     | Bug triage only                    |
| <b>Integration</b>   | Build the mock test site — ships today, blocks everyone | First end-to-end smoke test, however broken                  | Privacy inspector / dashboard v1                     | Lock demo script + record a backup video            | Final PPT + submit before deadline |

Day 4 is a mandatory full-team sync — this is where six parallel 3-day builds either click together or reveal a broken assumption while there's still one day to fix it. Don't skip it to keep coding.

# 1 Extension & Execution

**Owens this sprint:** Manifest V3 skeleton, content scripts, background service worker, action executor, client action validator.

Role 1 of 6 · EXTENSION · 5-day sprint

## HW Hardware

None needed. Any laptop, all 5 days.

## DAY Day-by-day

### Fri 28 Aug

- Set up the extension/ skeleton — manifest.json, content script, background service-worker stub.
- Get a “hello world” content script injecting into the mock site the moment Integration has it up.
- Agree the message-passing contract with Perception and Privacy Guard.

### Sat 29 Aug

- Build the action executor: map click/type/scroll to real browser APIs.
- Build the client action validator — target exists, action type allowed, confidence threshold met.

### Sun 30 Aug

- Build the type\_secret pattern — server requests a semantic op, real value resolved locally.
- Add basic error handling so a bad server response doesn't crash the extension mid-demo.

### Mon 31 Aug

- Full-team integration sync — plug into the live pipeline and fix whatever breaks.
- Help PVM/Verification wire the recovery path if actions fail.

### Tue 1 Sep

- Bug triage only. No new features today — stability over scope.

## DEP Dependencies

|                |                                                                                                       |
|----------------|-------------------------------------------------------------------------------------------------------|
| YOU DEPEND ON  | Server AI (action schema), Privacy Guard (nothing blocks you Day 1–2).                                |
| DEPENDS ON YOU | Everyone — you're the runtime host. If Day 1's skeleton slips, nobody else has anything to plug into. |

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## Perception & Local Inference

**Owens this sprint:** DOM/A11y capture and page-state normalization for this sprint. Local vision model (ONNX/WebGPU) is deferred past Sept 1.

Role 2 of 6 · PERCEPTION · 5-day sprint

### HW Hardware

None needed this sprint — DOM-only scope means no GPU work until after the deadline.

### DAY Day-by-day

#### Fri 28 Aug

- DOM/accessibility capture: element roles, labels, ARIA, form types.
- Normalize captured state into one structured representation everyone else can consume.

#### Sat 29 Aug

- Hand off the normalized state format to Privacy Guard (for detection) and Server AI (for the sanitized payload).

#### Sun 30 Aug

- Polish capture against the mock site's actual fields — this is where real bugs hide, not in generic DOM parsing.

#### Mon 31 Aug

- Full-team integration sync.
- Stress-test capture on the golden-path task, repeatably.

#### Tue 1 Sep

- Bug triage only.

### DEP Dependencies

|                |                                                                                                                            |
|----------------|----------------------------------------------------------------------------------------------------------------------------|
| YOU DEPEND ON  | Extension (content-script scaffold), Integration (mock site).                                                              |
| DEPENDS ON YOU | Privacy Guard and Server AI both consume your normalized state directly — a format change after Day 1 breaks both of them. |

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## Privacy Guard & Redaction

***Owens this sprint:** Tier-1 DOM-rule PII detection, deterministic redaction, redaction validation, the canary-value proof.*

Role 3 of 6 · PRIVACY · 5-day sprint

### HW Hardware

None needed. Any laptop, all 5 days.

### DAY Day-by-day

#### Fri 28 Aug

- Tier-1 DOM-rule detection only for this sprint (input type/autocomplete/labels/ARIA) — skip the text-NER and visual tiers for now.
- Deterministic redaction renderer: blackout/mask before anything is sent.

#### Sat 29 Aug

- Redaction validator — confirm every flagged field is actually handled.
- Build the sanitized-payload builder that Server AI's endpoint will consume.

#### Sun 30 Aug

- Design and build the canary-value test harness (e.g. CANARY\_EMAIL\_12345).
- Privacy Firewall: block the request outright on validation failure — this is your highest-value single feature for the demo.

#### Mon 31 Aug

- Full-team integration sync.
- Run the canary test end-to-end with Server AI; capture the proof artifact (payload log / hash).

#### Tue 1 Sep

- Bug triage only.

### DEP Dependencies

|                |                                                                                                                            |
|----------------|----------------------------------------------------------------------------------------------------------------------------|
| YOU DEPEND ON  | Perception (DOM state to run detection on), Extension (network-intercept point).                                           |
| DEPENDS ON YOU | Server AI must only ever receive your sanitized output — this is the single most judge-visible claim in the whole project. |

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Server AI

***Owens this sprint:** FastAPI + Pydantic, the cloud LLM/VLM call, the frozen action schema, context validation.*

Role 4 of 6 · SERVER · 5-day sprint

**HW Hardware**

First-claim priority on a laptop this sprint (fastest iteration on the live API integration) — but this still doesn't need a GPU, just a stable connection.

**DAY Day-by-day**

**Fri 28 Aug**

- FastAPI skeleton with a sanitized-context endpoint.
- Freeze the structured action schema (click/type/scroll + type\_secret) with Extension **today** — this blocks their Day 2 work.

**Sat 29 Aug**

- Get the endpoint live end-to-end, even if it returns a hardcoded action at first — unblocking Extension matters more than real reasoning on Day 2.

**Sun 30 Aug**

- Wire the real cloud LLM/VLM call, reasoning over the sanitized context and returning valid schema-conformant actions.

**Mon 31 Aug**

- Full-team integration sync.
- Run the canary proof from the server side — confirm the request log shows zero raw PII received.

**Tue 1 Sep**

- Bug triage only.

**DEP Dependencies**

|                |                                                                                                                   |
|----------------|-------------------------------------------------------------------------------------------------------------------|
| YOU DEPEND ON  | Privacy Guard (sanitized context format), Extension (the action schema, agreed Day 1).                            |
| DEPENDS ON YOU | Extension needs your response format to build its validator — late here means Extension is blocked, not just you. |

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## PVM & Verification

**Owens this sprint:** For this sprint: Level-1 deterministic verification and a basic recovery loop. Full PVM memory/caching is deferred past Sept 1.

Role 5 of 6 · PVM · 5-day sprint

### HW Hardware

None needed. Any laptop, all 5 days.

### DAY Day-by-day

#### Fri 28 Aug

- No verification target exists yet — spend today helping Integration build the mock site instead.

#### Sat 29 Aug

- Build Level-1 deterministic verification only: did the URL/DOM change the way the action predicted?

#### Sun 30 Aug

- Basic recovery loop: on a failed/ambiguous action, capture new state and retry once, safely.

#### Mon 31 Aug

- Full-team integration sync.
- Stress-test the one golden-path task repeatedly — this is where reliability either holds or doesn't.

#### Tue 1 Sep

- Bug triage only.

### DEP Dependencies

|                |                                                                                           |
|----------------|-------------------------------------------------------------------------------------------|
| YOU DEPEND ON  | Extension (action + result state to verify), Server AI (a real action to verify against). |
| DEPENDS ON YOU | Extension's recovery path hooks into your verification result.                            |

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## Integration, Benchmark & Demo

**Owens this sprint:** *The mock test site, end-to-end wiring, the privacy inspector v1, the demo script — full benchmark suite is deferred past Sept 1.*

Role 6 of 6 · INTEGRATION · 5-day sprint

### HW Hardware

Second-claim priority on a laptop for wiring/testing sessions.

### DAY Day-by-day

#### Fri 28 Aug

- Build the mock e-commerce/booking test site and ship it **today** — every other role is blocked without it.

#### Sat 29 Aug

- Run the first end-to-end smoke test across whatever pieces exist, however broken — the goal is finding the real integration gaps early.

#### Sun 30 Aug

- Build the privacy inspector / dashboard v1: redacted items, allow/block status. No sensitive values ever shown.

#### Mon 31 Aug

- Run and lead the full-team integration sync.
- Lock the demo script; record a backup video in case the live demo fails.

#### Tue 1 Sep

- Final PPT with real screenshots from the working prototype, not mockups. Submit before the deadline.

### DEP Dependencies

|                |                                                                                                                       |
|----------------|-----------------------------------------------------------------------------------------------------------------------|
| YOU DEPEND ON  | Every other role landing their Day 1–3 pieces on schedule.                                                            |
| DEPENDS ON YOU | Everyone — the mock site is the Day-1 blocker, and Day 4's sync is where six separate builds become one working demo. |